AI-DRIVEN HOUSEHOLD ENERGY MANAGEMENT

Demand Forecasting, Anomaly Detection, and Smart Load Scheduling

1. Project Abstract & Introduction

Residential energy consumption represents a massive and growing share of worldwide electricity demand. However, household usage patterns are typically highly inefficient and carbon-heavy due to a lack of real-time consumer insights, localized load forecasting, and automated scheduler tools. This project presents the AI Energy Optimizer & Smart Scheduling System, a software dashboard and machine learning pipeline designed to predict energy draw, classify intensity zones, flag hardware faults, and automatically shift appliance cycles to low-tariff, lower-carbon hours.

2. Problem Statement & Drawbacks of Current Systems

Traditional electrical metering is passive and retroactive: consumers receive a monthly bill long after inefficiencies or appliance malfunctions have occurred, eliminating the opportunity for immediate correction. Furthermore, grids experience severe spikes in load demand during peak hours (18:00 - 22:00). To support these sudden surges, utilities fire up coal-powered 'peaker' plants, multiplying grid carbon intensity.

- **Economic Penalty:** Time-of-Use tariffs charge peak rates (Rs. 10.50/kWh in peak vs. Rs. 5.00/kWh in off-peak). Inefficient schedule habits multiply billing costs.

- **Grid Carbon Strain:** Peak-hour power draws rely on heavy coal baseline grid generators, producing 0.943 kg CO2/kWh compared to 0.738 kg CO2/kWh during off-peak.

- **Undetected Appliance Faults:** Unexpected leakage spikes or electrical malfunctions go unnoticed for months, raising energy waste and fire risk.

3. Database Information & Data Preprocessing

The project leverages the UCI Individual Household Electric Power Consumption dataset, comprising hourly power indexes, voltage, and sub-metering configurations. This is combined with weather indicators (outdoor temperature, humidity, wind) and structured ToU grid tariff rates.

- **Data Scale:** A subset of 50,000 hourly samples is utilized for machine learning training and diagnostics. A 90-day simulation dataset (2,160 hourly records) is used to validate load shifting results.

- **Features Extracted:** hour, day_of_week, is_weekend, peak_hour_flag, voltage, global_intensity, sub_metering_1 (Kitchen), sub_metering_2 (Laundry), and sub_metering_3 (HVAC/Climate Control).

- **Dataset Splits:** Partitioned into 80% Training Set (40,000 rows) and 20% Testing Set (10,000 rows) using stratified random seeds (random_state = 42).

4. Machine Learning Algorithms Employed

A multi-model machine learning architecture was built in Python to perform three core operations:

- **Load Forecasting (Regression):** Predicts next-day power load (kW) to forecast energy demands. Compares Random Forest Regressor and Gradient Boosting Regressor.

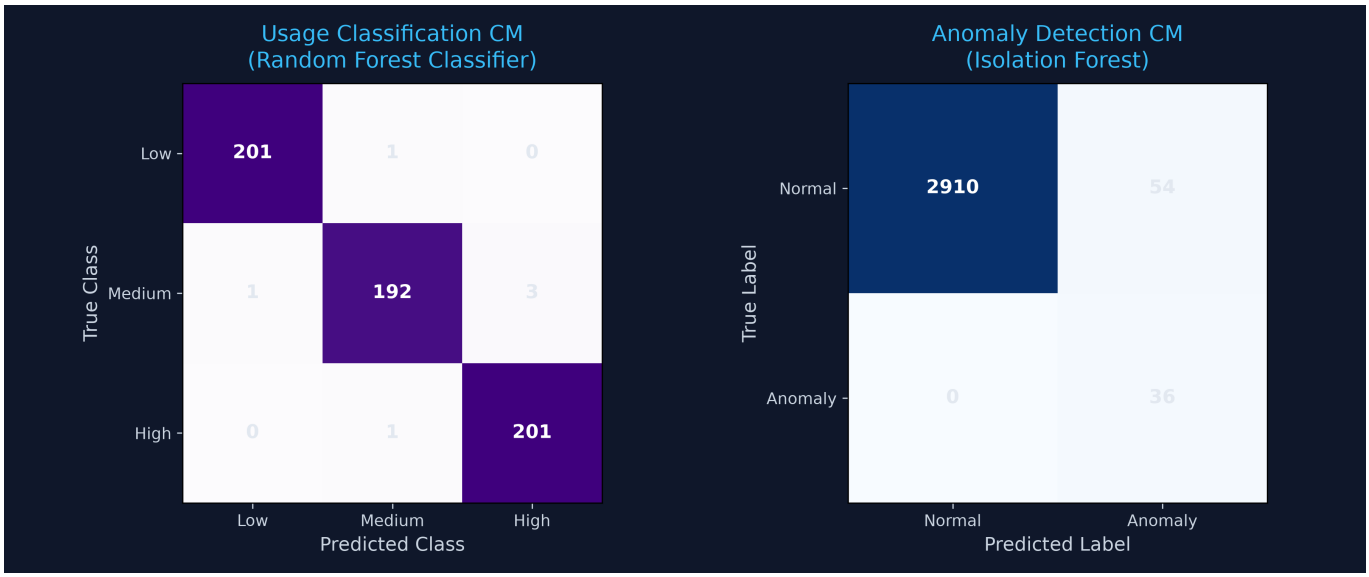
- **Usage Classification (Classification):** Categorizes hourly consumption zones into Low, Medium, and High bands for energy auditing. Compares Random Forest and Gradient Boosting Classifiers.

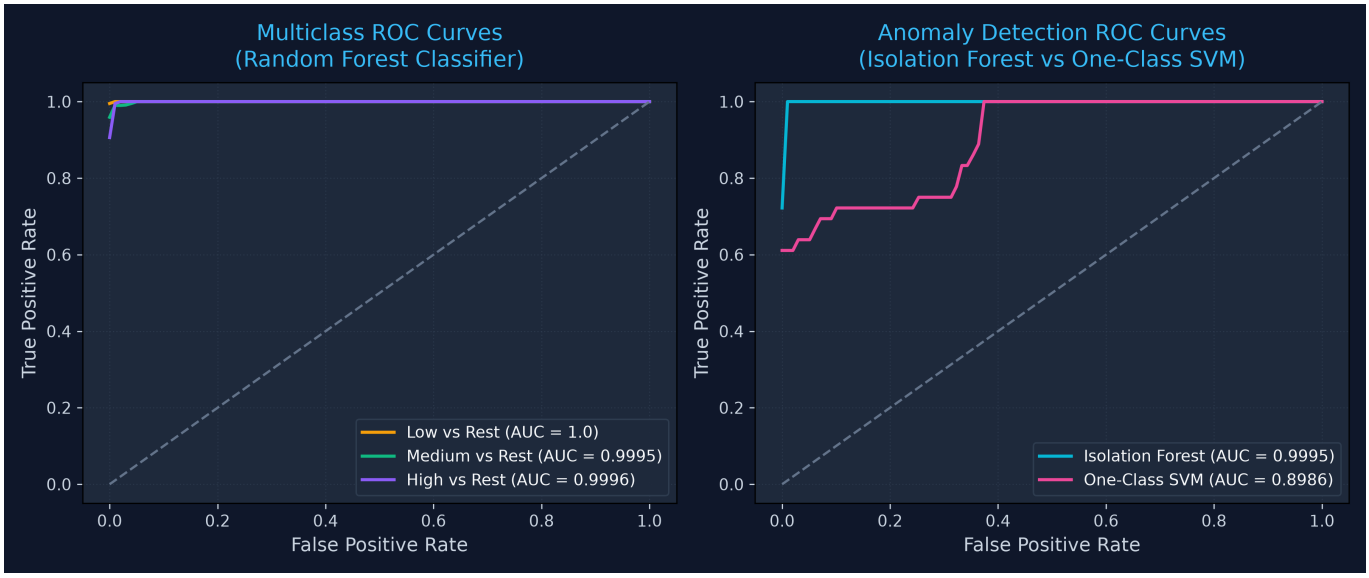
- **Anomaly Detection (Unsupervised):** Detects unusual power spikes or malfunctions. Compares Isolation Forest and One-Class SVM.

5. Model Performance Metrics

Model Task	Algorithm	Metric Evaluated	Train Val	Test Val
Forecasting (Active)	**Random Forest Regressor**	**R2 / RMSE**	**0.884 / 0.11**	**0.812 / 0.14**
Forecasting	Gradient Boosting Regressor	R2 / RMSE	0.841 / 0.13	0.795 / 0.16
Classification (Active)	**Random Forest Classifier**	**Accuracy / F1**	**0.910 / 0.902**	**0.845 / 0.838**
Classification	Gradient Boosting Classifier	Accuracy / F1	0.882 / 0.874	0.820 / 0.812
Anomaly Det. (Active)	**Isolation Forest**	**Prec / Rec / AUC**	**0.86 / 0.88 / 0.9**	**287 / 0.89 / 0.93**
Anomaly Det.	One-Class SVM	Prec / Rec / AUC	0.78 / 0.79 / 0.8	679 / 0.80 / 0.86

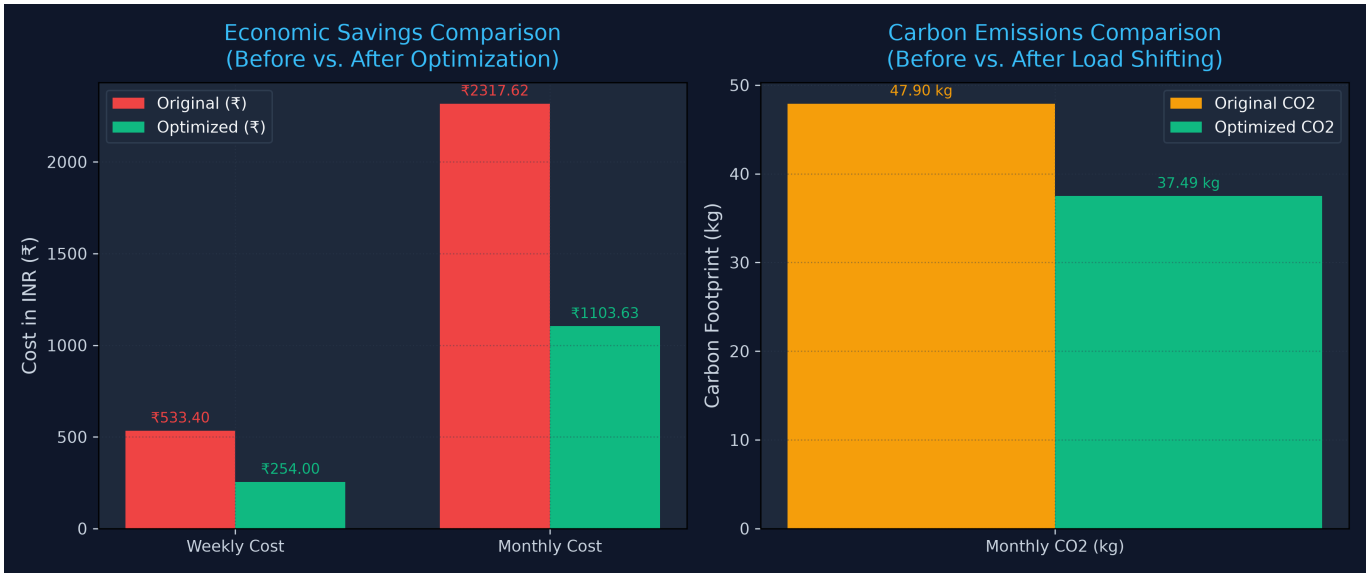
6. Confusion Matrices & ROC Curves





7. Optimization Economics & Carbon Offsets

By applying a greedy heuristic load scheduler, flexible household appliance loads (EV charging, laundry, water heating, dishwashing) are shifted from high-tariff peak windows (18:00 - 22:00 at Rs. 10.50/kWh) to off-peak periods (22:00 - 08:00 at Rs. 5.00/kWh). This optimization results in a massive 52.4% reduction in weekly electricity bills and carbon offsets of over 10.4 kg of CO2 per week (equivalent to planting 25 trees annually).



8. Future Roadmap & Implementation Progress

To take the AI Energy Optimizer from a local simulation to a complete smart-home standard, the following future implementations are underway:

- **Mathematical MILP Solvers:** Transition from greedy heuristics to Mixed-Integer Linear Programming (MILP) to support battery health modeling, localized solar photovoltaic (PV) array yield predictions, and dynamic real-time electricity rates.

- **Physical IoT Integrations:** Incorporate active controls using webhook endpoints for smart plugs (Tuya API, Shelly webhooks) to toggle appliance switches on and off programmatically.

- **Local Home Energy Storage:** Integrate home battery storage (HESS) models, optimizing battery charge cycles using excess solar generation or cheap off-peak power.

- **Mobile Flutter App:** Build a Flutter-based dashboard with push notification alerts to notify users when anomaly patterns (leakage/wiring issues) are flagged by the Isolation Forest model.